

Requirement & Design Specification

Global Access Management System (GAMS)

Version: 1.0

– Hanoi, August 2022 –

Record of Changes

Version	Date	A* M, D	In charge	Change Description
V1.0	15/2	A	KienNTHE11	

*A - Added M - Modified D - Deleted

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I. Overview

1. User Requirements

1.1 Actors

[An actor is a person (or sometimes another software system or a hardware device) that interacts with the system to perform a use case. Following are some questions you might ask to help user representatives identify actors

- *Who (or what) is notified when something occurs within the system?*
- *Who (or what) provides information or services to the system?*
- *Who (or what) helps the system respond to and complete a task?*

This part gives the description of system actors, you can follow the table form as below]

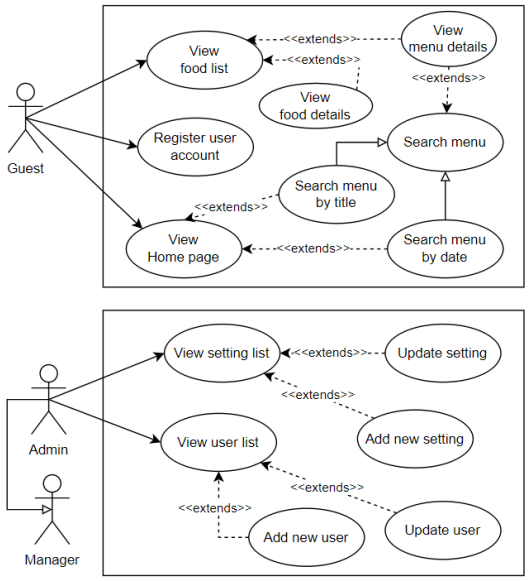
#	Actor	Description
1	Admin	...
2	...	...

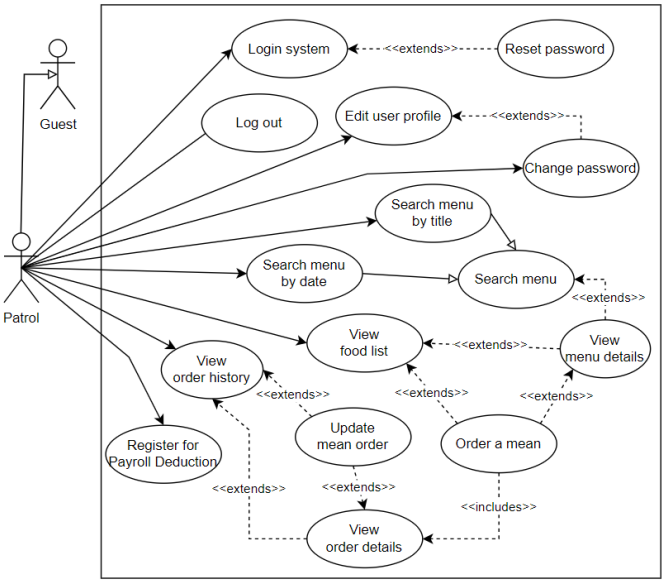
1.2 Use Cases

[A use case (UC) describes a sequence of interactions between a system and an external actor that results in the actor being able to achieve some outcome of value. The names of use cases are always written in the form of a verb followed by an object. Select strong, descriptive names to make it evident from the name that the use case will deliver something valuable for some user]

a. Diagram(s)

[Provide the UC diagram(s) to show the actor-UCs and UC-UC relationships like the sample below. You can have multiple UC diagrams for the system]





b. Descriptions

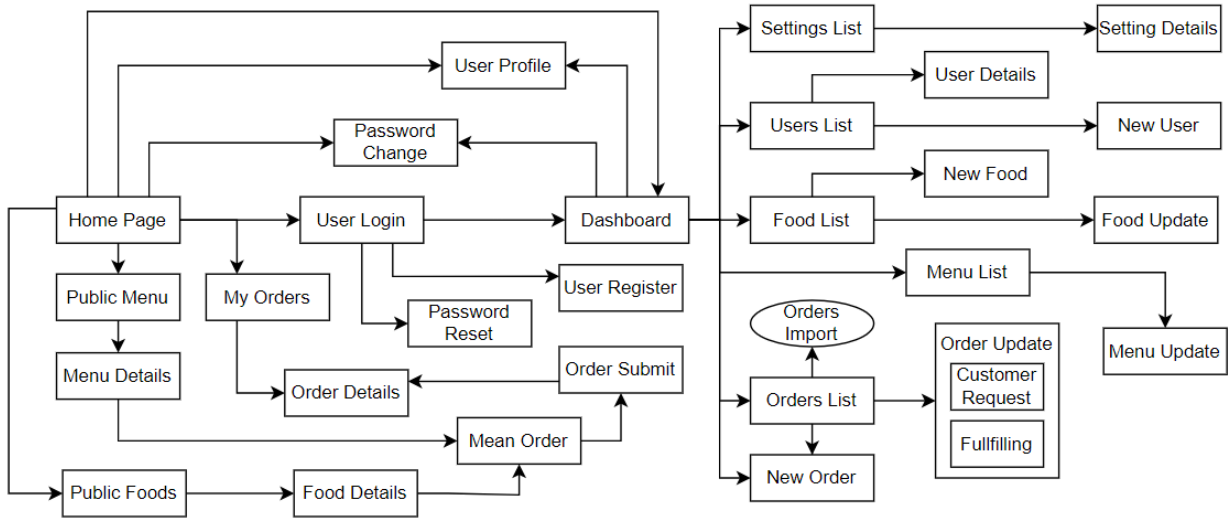
This part describes the use cases, you can follow the table form as below]

ID	Feature	Use Case	Use Case Description
01	Menu Operations	View Menu	…
02	Order Meals	Order a Meal	…
03	…	…	…

2. Overall Functionalities

2.1 Screens Flow

[This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below. Please note that beside the normal flat screen, we might have the oval notation for pop-up screen (Import Order) or a screen with multiple information tab (Order Details), etc. You may also use text or background format for different visuality purpose]



2.2 Screen Descriptions

[Provide the descriptions for the screens in the Screens Flow above]

#	Feature	Screen	Description
1	Order Meals	Create Order	<<Screen Brief description>>
2	Order Meals	Change Order	
3	..		

2.3 Screen Authorization

[Provide the system roles authorization to the system features (down to screens, and event to the screen activities if applicable) in the table form as below – replace Role1, Role2,… with your specific system user role names]

Screen\Function	Role-Name1	Role-Name2	Role-Name3	…
<<Screen Name1>>	X		X	X
<<Screen Activity>>			X	X
<<Screen Name2>>	X		X	
Query All Data	X			
Query Own Data			X	
Query Managed Data			X	
Add New Data			X	X
Update All Data				X
Update Own Data				X
Update Managed Data				X
Delete Data				

…				

2.4 Non-UI Functions

[Provide the descriptions for the functions which have no UI (or not screens), i.e batch/cron job, service, API, etc.]

#	Feature	System Function	Description
1	<<Feature Name>>	<<Function Name1>>	<<Function Name1 Description>>
2	…		

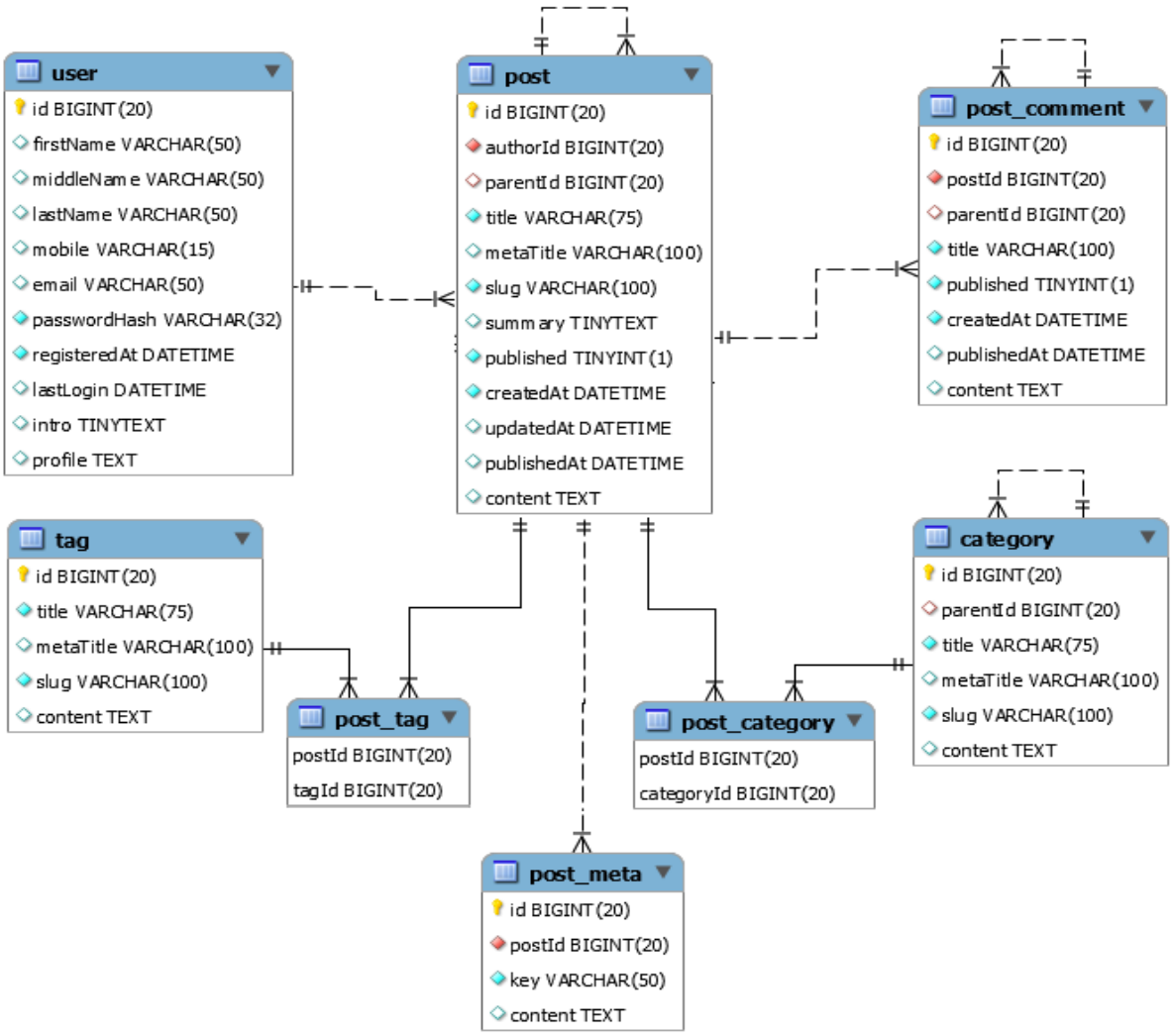
3. System High Level Design

3.1 ERD

3.2. Database Design

[Provide the tables relationship like example below]

a. Database Schema

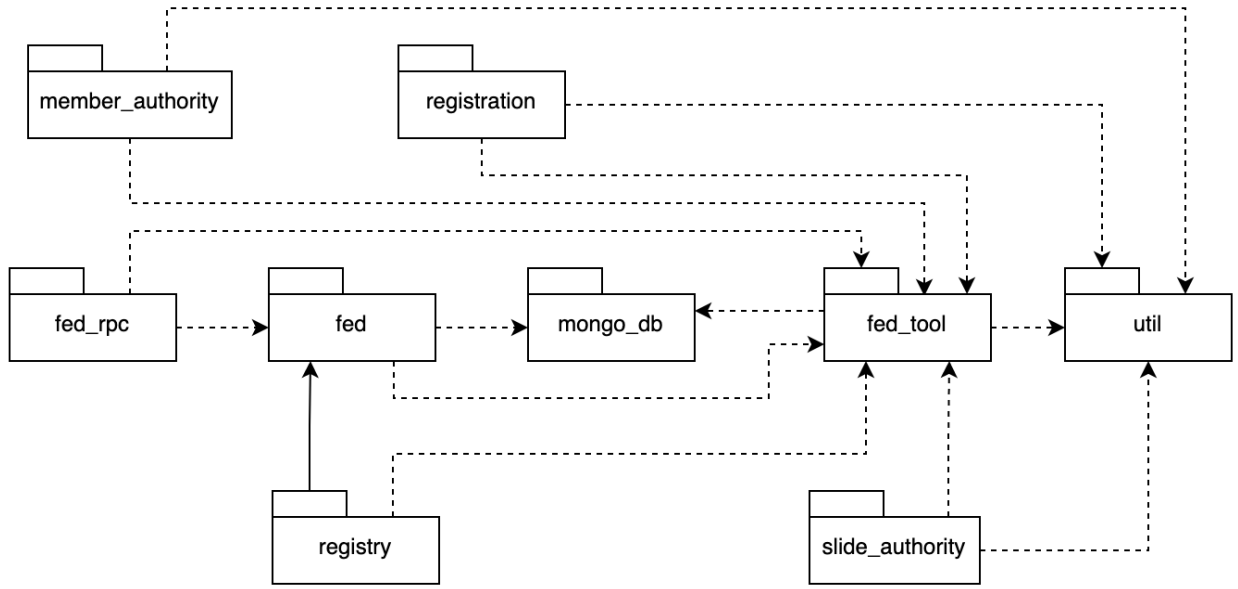


b. Table Descriptions

No	Table	Description
01	*<Table name>*	*<Description of the table>* *- Primary keys: <<list of primary key fields>>* *- Foreign keys: <<list of foreign key fields>>*
02	*<Table name2>*	...

3.2 Code Packages

[Provide the package diagram for each sub-system. The content of this section including the overall package diagram, the explanation, package and class naming conventions in each package. Please see the sample & description table format below]



Package descriptions

No	Package	Description
01	*Member_authority*	*<Description of the package>*
02	*registration*	*<Description of the package>*
03	*...*	

II. Requirement Specifications

1. <<Feature Name>>

1.1 <<UseCaseCode_UC Name>>

a. Functionalities

Provide the functional description for the use cases using the template/guides below

Functional Description Template

UC ID and Name:			
Created By:		Date Created:	
Primary Actor:		Secondary Actors:	
Trigger:			
Description:			
Preconditions:			
Postconditions:			
Normal Flow:			
Alternative Flows:			
Exceptions:			
Priority:	High (Medium, Low), Must Have (Should Have, Could Have),..		
Frequency of Use:			
Business Rules:			
Other Information:			
Assumptions:			

Functional Description Contents

Use Case ID and Name

Give each use case a unique integer sequence number identifier. State a concise name for the use case that indicates the value the use case would provide to some user. Begin with an action verb, followed by an object.

Author and Date Created

Enter the name of the person who initially wrote this use case and the date it was written.

Primary and Secondary Actors

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the primary actor that will be initiating this use case and any other secondary actors who will participate in completing execution of the use case.

Trigger

Identify the business event, system event, or user action that initiates the use case. This trigger alerts the system that it should begin testing the preconditions for the use case so it can judge whether to proceed with execution.

Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. The system must be able to test each precondition. Number each precondition. Example: PRE-1: User's identity has been authenticated.

Postconditions

Describe the state of the system at the successful conclusion of the use case execution. Label each postcondition in the form POST-X, where X is a sequence number. Example: POST-1: Price of item in the database has been updated with the new value.

Normal Flow

Provide a description of the user actions and corresponding system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. Show a numbered list of actions performed by the actor, alternating with responses provided by the system. The normal flow is numbered "X.0", where "X" is the Use Case ID.

Alternative Flows

Document other successful usage scenarios that can take place within this use case. State the alternative flow, and describe any differences in the sequence of steps that take place. Number each alternative flow in the form "X.Y", where "X" is the Use Case ID and Y is a sequence number for the alternative flow. For example, "5.3" would indicate the third alternative flow for use case number 5. Indicate where each alternative flow would branch off from the normal flow, and if pertinent, where it would rejoin the normal flow.

Exceptions

Describe any anticipated error conditions that could occur during execution of the use case and how the system is to respond to those conditions. Number each alternative flow in the form "X.Y.EZ", where "X" is the Use Case ID, Y indicates the normal (0) or alternative (>0) flow during which this exception could take place, "E" indicates an exception, and "Z" is a sequence number for the exceptions. For example "5.0.E2" would indicate the second exception for the normal flow for use case number 5. Indicate where in the normal (or an alternative) flow each exception could occur.

Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. Use the same priority scheme as that used for the functional requirements.

Frequency of Use

Estimate the number of times this use case will be performed per some appropriate unit of time. This gives an early indicator of throughput, concurrent usage loads, and transaction capacity.

Business Rules

List any business rules that influence this use case. Don't include the business rule text here, just its identifier so the reader can find it in another repository when needed.

Other Information

Identify any additional requirements, such as quality attributes, for the use case that may need to be addressed during design or implementation. Also list any associated functional requirements that aren't a direct part of the use case flows but which a developer needs to know about. Describe what should happen if the use case execution fails for some unanticipated or systemic reason (e.g., loss of network connectivity, timeout). If the use case results in a durable state change in a database or the outside world, state whether the change is rolled back, completed correctly, partially completed with a known state, or left in an undetermined state as a result of the exception.

Assumptions

List any assumptions that were made regarding this use case or how it might execute.

b. Business Rules

Provide the business rules those are applied only to the use case

ID	Business Rule	Business Rule Description
FR1	Password Encoding	User's password must be encoded with MD5 hashing

2. Common Functions

2.1 UC-2_Login System

a. Functional Description

UC ID and Name:	**UC-2_Login System**		
Created By:	MinhNNT	Date Created:	16/Jun/2023
Primary Actor:	Customer	Secondary Actors:	None
Trigger:	User clicks Login button from the page header, or User accesses an authenticated feature (from a link or type the page URL directly into the address bar)		
Description:	As a user, I want to be able to log into the system so that I can use the system's authenticated features and access my personalized account.		
Preconditions:	User account has been created & authorized		
Postconditions:	• User logs in the system successfully • The system tracked successful login into the Activity Log		
Normal Flow	**2.0 Login System** 1. User accesses the User Login screen 2. User types in the login details or choo other login options (see 2.1 and 2.2) 3. User clicks the Login button 4. System validates the login details (see 2.0.E1) 5. System allows user to access 6. System tracks user's success login to the Activity Log 7. System accesses the Home Page (or the previous calling page if any)		
Alternative Flows:	***2.1 Google Login*** 1. User chooses to login system using Google account 2. System redirects the user to the Google's Login screen 3. User types in the Google account details and chooses to login 4. Google validates user's login information successfully and redirect him/her back to the system 5. Return to step 5 of normal flow. ***2.2 Facebook Login*** 1. User chooses to login system using Facebook account 2. System redirects the user to the Facebook's Login screen 3. User types in the Facebook account details and chooses to login 4. Facebook validates user's login information successfully and redirect him/her back to the system 5. Return to step 5 of normal flow.		
Exceptions:	***2.0.E1 System can't authenticate the user*** 1. The Error Message screen is shown to the user 2. User cancels the logging in *=> UC stops, change to UC-1_View Home Page* 3. User clicks "Forgot Password?" link *=> change to UC-3_Reset Password* 4. User clicks "Register" link *=> change to UC-4_Register User Account*		
Priority:	Must Have		
Frequency of Use:			
Business Rules:	FR1, FR2, FR3		

Other Information:	
Assumptions:	

b. Business Rules

ID	Business Rule	Business Rule Description
FR1	Password Encoding	User's password must be encoded with MD5 hashing
FR2	Invalid Logging In	User can't be authenticated to login the system if below cases • His/her logging-in details are incorrect • His/her account has not been verified • His/her account has been locked or blocked
FR3	Account Locking	If user inputs wrong logging-in details 6 times continuously, his/her account would be locked in 30 minutes

3. Patron Feature

3.1 UC-5_Order a Meal

a. Functional Description

ID and Name:	**UC-5 Order a Meal**		
Created By:	Prithvi Raj	Date Created:	10/4/13
Primary Actor:	Patron	Secondary Actors:	Cafeteria Inventory System
Description:	A Patron accesses the Cafeteria Ordering System from the corporate intranet or from home, views the menu for a specific date if desired, selects food items, and places an order for a meal to be delivered to a specified location within a specified 15-minute time window.		
Trigger:	A Patron indicates that he wants to order a meal		
Preconditions:	PRE-1. Patron is logged into COS. PRE-2. Patron is registered for meal payments by payroll deduction.		
Postconditions:	POST-1. Meal order is stored in COS with a status of "Accepted". POST-2. Inventory of available food items is updated to reflect items in this order. POST-3. Remaining delivery capacity for the requested time window is updated.		
Normal Flow:	**5.0 Order a Single Meal** 1. Patron asks to view menu for a specific date. (see 5.0.E1, 5.0.E2) 2. COS displays menu of available food items and the daily special. 3. Patron selects one or more food items from menu. (see 5.1) 4. Patron indicates that meal order is complete. (see 5.2) 5. COS displays ordered menu items, individual prices, and total price, including taxes and delivery charge. 6. Patron either confirms meal order (continue normal flow) or requests to modify meal order (return to step 2). 7. COS displays available delivery times for the delivery date. 8. Patron selects a delivery time and specifies the delivery location. 9. Patron specifies payment method. 10. COS confirms acceptance of the order. 11. COS sends Patron an email message confirming order details, price, and delivery instructions.		

	12. COS stores order, sends food item information to Cafeteria Inventory System, and updates available delivery times.
Alternative Flows:	**5.1 Order multiple identical meals** 1. Patron requests a specified number of identical meals. (see 5.1.E1) 2. Return to step 4 of normal flow. **5.2 Order multiple meals** 1. Patron asks to order another meal. 2. Return to step 1 of normal flow.
Exceptions:	**5.0.E1 Requested date is today and current time is after today's order cutoff time** 1. COS informs Patron that it's too late to place an order for today. 2a. If Patron cancels the meal ordering process, then COS terminates use case. 2b. Else if Patron requests another date, then COS restarts use case. **5.0.E2 No delivery times left** 1. COS informs Patron that no delivery times are available for the meal date. 2a. If Patron cancels the meal ordering process, then COS terminates use case. 2b. Else if Patron requests to pick the order up at the cafeteria, then continue with normal flow, but skip steps 7 and 8. **5.1.E1 Insufficient inventory to fulfill multiple meal order** 1. COS informs Patron of the maximum number of identical meals he can order, based on current available inventory. 2a. If Patron modifies number of meals ordered, then Return to step 4 of normal flow. 2b. Else if Patron cancels the meal ordering process, then COS terminates use case.
Priority:	High
Frequency of Use:	Approximately 300 users, average of one usage per day. Peak usage load for this use case is between 9:00 A.M. and 10:00 A.M. local time.
Business Rules:	BR-1, BR-2, BR-3, BR-4, BR-11, BR-12, BR-33
Other Information:	1. Patron shall be able to cancel the meal ordering process at any time prior to confirming it. 2. Patron shall be able to view all meals he ordered within the previous six months and repeat one of those meals as the new order, provided that all food items are available on the menu for the requested delivery date. (Priority = M) 3. The default date is the current date if the Patron is using the system before today's order cutoff time. Otherwise, the default date is the next day that the cafeteria is open.
Assumptions:	Assume that 15 percent of Patrons will order the daily special (source: previous 6 months of cafeteria data).

b. Business Rules

None

3.2 UC-6_Register for Payroll Deduction

a. Functional Description

ID and Name:	**UC-6 Register for Payroll Deduction**		
Created By:	Nancy Anderson	Date Created:	9/15/13
Primary Actor:	Patron	Secondary Actors:	Payroll System

Description:	Cafeteria patrons who use the COS and have meals delivered must be registered for payroll deduction. For noncash purchases made through the COS, the cafeteria will issue a payment request to the Payroll System, which will deduct the meal costs from the next scheduled employee payday direct deposit.
Trigger:	Patron requests to register for payroll deduction, or Patron says yes when COS asks if he wants to register
Preconditions:	PRE-1. Patron is logged into COS.
Postconditions:	POST-2. Patron is registered for payroll deduction.
Normal Flow:	**6.0 Register for Payroll Deduction** 1. COS asks Payroll System if Patron is eligible to register for payroll deduction. 2. Payroll System confirms that Patron is eligible to register for payroll deduction. 3. COS asks Patron to confirm his desire to register for payroll deduction. 4. If so, COS asks Payroll System to establish payroll deduction for Patron. 5. Payroll System confirms that payroll deduction is established. 6. COS informs Patron that payroll deduction is established.
Alternative Flows:	None
Exceptions:	6.0.E1 Patron is not eligible for payroll deduction 6.0.E2 Patron is already enrolled for payroll deduction
Priority:	High
Business Rules:	BR-86 and BR-88 govern an employee's eligibility to enroll for payroll deduction.
Other Information:	Expect high frequency of executing this use case within first 2 weeks after system is released.

b. Business Rules

None

III. Design Specifications

1. <<Feature Name>>

1.1 <<SubFeature Name>>

a. <<Screen/Function Name>>

[Provide brief description of the screen/function + related UC here and other details as in the sub-sections]

UI Design

[This is to describe the UI layout (Mockup prototype) & descriptions for screen fields/components]

<<Mockup prototype>>

Field Name	Field Type	Description
Field Group Name		
<<Field-Name>>	<<Field type>>	<<Field description & data initializing design>>

Database Access

[Provide the design description for the screen/function to access the database here: what table the screen/function would access, which transactions does it make (C-Create, R-Read, U-Update, or D-Delete), and how/purpose of the access (by providing Description and SQL commands)]

Table	CRUD	Description
<<Table Name>>	<<transaction(s)>>	<<Table access description: purpose, how,…>>
..		

SQL Commands

[Provide the detailed SQL (select, insert, update...) which are used in implementing the screen/function]

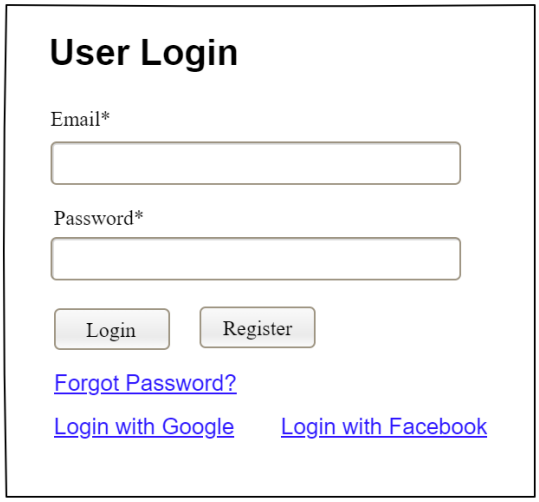
1.2 System Access

a. User Login

This screen allows user to be authenticated to the system screens/functionalities.
Related use cases:

- UC02_Login System

UI Design



Field Name	Field Type	Description
Email*	Text Box	This is for user to input valid email address for logging in
Password*	Password Box	This is for user to input password for logging in
Login	Button	User clicks to authenticate him/herself into the system with provided email & password
Register	Button	User clicks to redirect to the User Register page for registering new user account to access the system
Forgot Password?	Hyperlink	User clicks to redirect to the Password Reset page for resetting his/her forgot password
Login with Google	Hyperlink	Allow user to login with his/her Google account
Login with Facebook	Hyperlink	Allow user to login with his/her Facebook account

Database Access

Table	CRUD	Description
User	R	Verify UserName & Password information
Setting, User	R	Specify the authorizations of the logged-in user

SQL Commands:

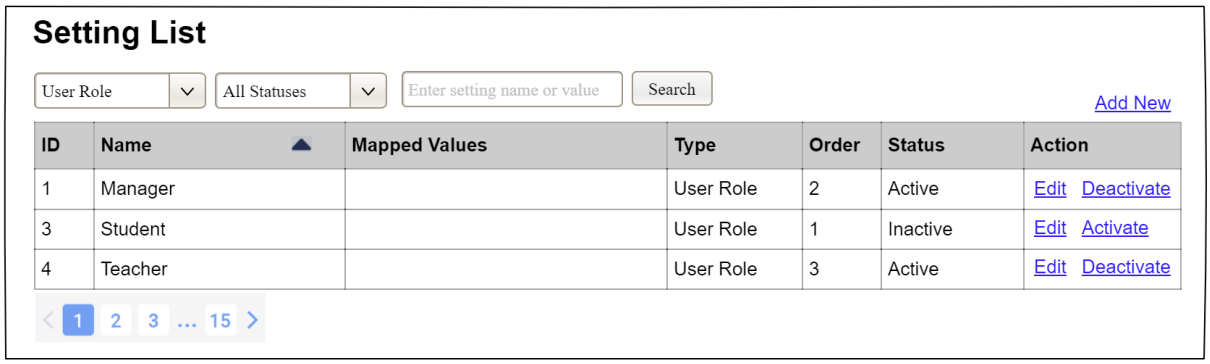
1/ Verify UserName & Password information

```
SELECT user_id, full_name, email, image_url, status
FROM user WHERE user_name = ? AND password = ?
```

2/ Specify the authorizations of the logged-in user
SELECT mapped_values FROM setting WHERE setting_id = ?
SELECT setting_name, mapped_values FROM setting WHERE setting_id IN (?)

b. Setting List

UI Design



Field Name	Field Type	Description
Search Fields		
Setting Type	Combo Box Single-Choice	Filled with the list of current active setting types Allow to filter the list by setting type; Default value is “All Types”
Setting Status	Combo Box Single-Choice	Values: All Statuses (default), Active, and Inactive Allow to filter the list by status Default value: “All Statuses”
Search Phase	Text Box String (30)	Allow to search using the name or map values Default value: blank
Search	Button	Click to refresh the list with the defined filter(s) and search phrase.
Add New	Hyperlink	Click to open the Setting Details page for adding new setting (master data)
Data Table		
ID	Integer	Auto-increased identifier of the setting
Name	Text	Name of the setting
Mapped Values	Text	Supplementary information for the setting
Type	Text	Type of the setting
Order	Integer	Display order of the setting: the order of the setting type, displayed among the list of settings with the same type
Data Actions		
Edit	icon	Click to open the Setting Details page for updating the relevant setting (master data)
Activate	icon	Shown when the data status is inactive. This is to activate the relevant setting (master data)
Deactivate	Ion	Shown when the data status is active. This is to deactivate the relevant setting (master data)

Database Access

Table	CRUD	Description
Setting	RU	Query the list of current settings from the database Update status of a specific setting

SQL Commands:

1/ Query the list of current settings from the database

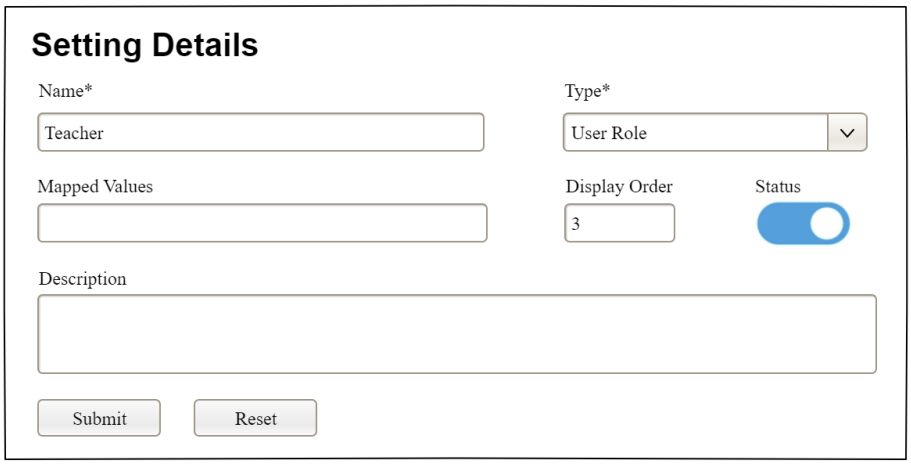
```
SELECT setting_id, setting_name, mapped_values, type_id, display_order, status
FROM setting WHERE (setting_type = ?) AND (status = ?) AND  (setting_name LIKE ?)
```

2/ Update status of a specific setting

```
UPDATE setting SET status = ? WHERE setting_id = ?
```

c. Setting Details

UI Design



Field Name	Field Type	Description
Name*	Text Box String (20)	Name of the setting
Type*	Combo Box (Single Choice)	Type of the setting, filled with the list of setting types Default value: the first type in the list
Mapped Values	Text Box String (50)	Supplementary information for the setting (if any)
Order	Text Box Integer (>=0)	Display order of the setting: the order of the setting type, displayed among the list of settings with the same type
Status	On/Off button	Status of the setting: Active or Inactive Default value: Active
Description	Text Area String (200)	Description of the setting
Submit	Button	Click to store new or updated setting details
Reset	Button	Click to reset the changes use has made on the screen fields back to the initial values when the screen is loaded

Database Access

…

IV. Appendix

1. Assumptions & Dependencies

[Record any assumptions that were made when conceiving the project and writing this vision and scope document. Note any major dependencies the project must rely upon for success, such as specific technologies, third-party vendors, development partners, or other business relationships.]

<<Sample:

AS-1: Systems with appropriate user interfaces will be available for cafeteria employees to process the expected volume of meals ordered.

AS-2: Cafeteria staff and vehicles will be available to deliver all meals for specified delivery time slots within 15 minutes of the requested delivery time.

DE-1: If a restaurant has its own on-line ordering system, the Cafeteria Ordering System must be able to communicate with it bi-directionally.

>>

2. Limitations & Exclusions

[Identify any product features or characteristics that a stakeholder might anticipate, but which are not planned to be included in the new product]

3. Business Rules

[Provide common business rules that you must follow. The information can be provided in the table format as the sample below]

<<Sample

ID	Category	Rule Definition
BR-01	Constraints	Delivery time windows are 15 minutes, beginning on each quarter hour.
BR-02	Constraints	Deliveries must be completed between 10:00 A.M. and 2:00 P.M. local time, inclusive.
BR-03	Facts	All meals in a single order must be delivered to the same location.
BR-04	Facts	All meals in a single order must be paid for by using the same payment method.
BR-11	Constraints	If an order is to be delivered, the patron must pay by payroll deduction.
BR-12	Computations	Order price is calculated as the sum of each food item price times the quantity of that food item ordered, plus applicable sales tax, plus a delivery charge if a meal is delivered outside the free delivery zone.

>>

4. ..